

LESSON PLAN

Lesson Metadata

Week	1
Week Ending	11-09-2026
School	Achimota Basic School
Teacher	Persistence Teacher
Teaching Week	2
Lesson Date	14/09/2026
Duration (minutes)	60
Lesson Topic	Input devices: manual and automatic

Curriculum Alignment

Strand	Introduction to Computing
Sub Strand	Components of Computers and Computer Systems
Content Standard	B7.1.1.1
Learning Indicator	B7.1.1.1.1
Performance Indicator	Learners can discuss fourth-generation computers and the microchip
Core Competencies	Critical Thinking and Problem Solving, Communication and Collaboration, Digital Literacy
Key Words	manual and automatic input devices, Processor, fourth-generation computers, microchip, microprocessor, computer generations, components, computers

WAEF Special Fields

Through line	God worshiper and Image reflector
God's Story	Creation
Deep Hope	Learners will recognize and appreciate the beauty, order, and purpose of design in the physical world around them.
Storyline	Shaping our world.

Lesson Delivery

Phase 1 Starter	- Display pictures of a keyboard, mouse, touchscreen and barcode reader. Ask learners which they have used at home and record their answers in two columns: devices a person works and devices that work on their own. - Hold up a microchip (or a clear picture of one) next to a picture of an early room-sized computer. Ask learners which is newer and why, write their reasons on the board, then tell them both are computers.
Phase 2 Main	- Watch and listen carefully - then describe each step of Fourth-generation computers and the microchip in your own words - Work with your partner/group to carry out the task for Fourth-generation computers and the microchip and record what you find - Give each pair a labelled diagram of a keyboard and ask them to point to the home row - the function keys and the number pad - then explain which of the devices sends data to the computer automatically - Give pairs a labelled diagram of a processor. Learners trace the path from the microchip to the parts it controls and write one sentence on why the microchip changed computers - Demonstrate the concept or steps for Fourth-generation computers and the microchip. Use the board or an unplugged model if there are not enough devices for everyone - Class Assignment: In pairs, learners classify a set of device pictures as manual or automatic and write one advantage and one disadvantage of each group; the teacher checks the reasons given.
Phase 2 Resources	- Touchscreen (corrected spelling) - Processor - a real microchip or a clear picture of one - diagram of a processor - chart comparing computer generations
Phase 3 Plenary	- Learners state one manual and one automatic input device they used today and explain in one sentence why a hospital or a supermarket would choose the automatic one. - Home Assignment: List three input devices used in your home or community. For each one write whether it is manual or automatic, what it is used for and one care you must take when using it.
Evaluation	Give each pair four device cards and ask them to sort them as manual or automatic, giving one reason per card; ask two pairs to justify a borderline device and listen for the words manual and automatic used correctly.

Assignments

Class Assignment	In pairs, learners classify a set of device pictures as manual or automatic and write one advantage and one disadvantage of each group; the teacher checks the reasons given.
Home Assignment	List three input devices used in your home or community. For each one write whether it is manual or automatic, what it is used for and one care you must take when using it.
Homework	List three input devices used in your home or community. For each one write whether it is manual or automatic, what it is used for and one care you must take when using it.

References

Type	Title	Author / Publisher	Page
ICT Curriculum	Computing Curriculum, Basic 7, pages 12-14		

LESSON PLAN

Lesson Metadata

Week	2
Week Ending	18-09-2026
School	Achimota Basic School
Teacher	Persistence Teacher
Teaching Week	2
Lesson Date	16/09/2026
Duration (minutes)	60
Lesson Topic	Input devices: manual and automatic

Curriculum Alignment

Strand	Introduction to Computing
Sub Strand	Components of Computers and Computer Systems
Content Standard	B7.1.1.1
Learning Indicator	B7.1.1.1.2
Performance Indicator	Learners can demonstrate input devices: manual and automatic
Core Competencies	Critical Thinking and Problem Solving, Communication and Collaboration, Digital Literacy
Key Words	mouse, Personal Computer, keyboard, input devices, touchscreen, barcode reader, scanner, manual input, automatic input, components, computers

WAEF Special Fields

Lesson Delivery

Phase 1 Starter	- This lesson builds on 'Components of Computers and Computer Systems' and moves the class on to Input devices: manual and automatic. - Build on the previous lesson ('Components of Computers and Computer Systems'). Show pictures of a keyboard, mouse, touchscreen, barcode reader and scanner. Ask learners which of these they have used and record their answers on the board.
Phase 2 Main	- Watch and listen carefully - then describe each step of Input devices: manual and automatic in your own words - Work with your partner/group to carry out the task for Input devices: manual and automatic and record what you find - Demonstrate one device from each group in front of the class: type on the keyboard (manual) and scan a barcode (automatic). Learners watch and then say which needed a person and which worked on its own - Give each pair the device pictures. Learners sort them into manual and automatic groups - then list one advantage and one disadvantage of each group and say where each device is used - Demonstrate the concept or steps for Input devices: manual and automatic. Use the board or an unplugged model if there are not enough devices for everyone - Class Assignment: Pairs sort the device pictures into manual and automatic groups, write one reason for each group, and state where each device is used. The teacher checks a sample and asks two pairs to justify a borderline device.
Phase 2 Resources	- mouse - Personal Computer - keyboard - picture cards showing keyboard - touchscreen - barcode reader - scanner - a real keyboard and a real barcode reader or scanner - sorting chart labelled MANUAL / AUTOMATIC
Phase 3 Plenary	- Review the steps for Input devices: manual and automatic, discuss the common problem met and how it was solved, and link to everyday technology. Preview the next lesson ('Components of Computers and Computer Systems'). - Home Assignment: Find and list three input devices used in your home or community. For each one, say whether it is manual or automatic and what it is used for.
Evaluation	Give each pair four device cards and ask them to sort the cards as manual or automatic, giving one reason per card. Ask two pairs to explain a borderline device and listen for the words 'manual' and 'automatic' used correctly. Assessment mode: Oral questions and observation of participation.

Assignments

Class Assignment	Pairs sort the device pictures into manual and automatic groups, write one reason for each group, and state where each device is used. The teacher checks a sample and asks two pairs to justify a borderline device.
Home Assignment	Find and list three input devices used in your home or community. For each one, say whether it is manual or automatic and what it is used for.
Homework	Find and list three input devices used in your home or community. For each one, say whether it is manual or automatic and what it is used for.

LESSON PLAN

Lesson Metadata

Week	3
Week Ending	25-09-2026
School	Achimota Basic School
Teacher	Persistence Teacher
Teaching Week	3
Lesson Date	21/09/2026
Duration (minutes)	60
Lesson Topic	Hard drives and optical storage media

Curriculum Alignment

Strand	Introduction to Computing
Sub Strand	Components of Computers and Computer Systems
Content Standard	B7.1.1.1
Learning Indicator	B7.1.1.1.4
Performance Indicator	Learners can identify hard drives and optical storage media
Core Competencies	Critical Thinking and Problem Solving, Communication and Collaboration, Digital Literacy
Key Words	Pen drive, Hard Disk, hard drive, portable hard drive, optical disc, CD-ROM, DVD-ROM, storage capacity, write speed, disk caching, components, computers

WAEF Special Fields

Lesson Delivery

Phase 1 Starter	- This lesson builds on 'Components of Computers and Computer Systems' and moves the class on to Hard drives and optical storage media. - Build on the previous lesson ('Components of Computers and Computer Systems'). Display a portable hard drive and two optical discs (one CD, one DVD). Ask learners where they save school work and why, and note their answers on the board.
Phase 2 Main	- Watch and listen carefully - then describe each step of Hard drives and optical storage media in your own words - Work with your partner/group to carry out the task for Hard drives and optical storage media and record what you find - Name and point to each storage item: magnetic hard drive - portable hard drive - CD-ROM - DVD-ROM. Learners repeat each name and say which one holds the most data and why - In groups - learners read the printed capacities and write speeds from the disc labels into a table - then compare the items and state one strength and one weakness of each - Demonstrate the concept or steps for Hard drives and optical storage media. Use the board or an unplugged model if there are not enough devices for everyone - Class Assignment: Groups build a comparison table of the hard drive, portable hard drive, CD-ROM and DVD-ROM using the printed capacities and write speeds, then state one strength and one weakness of each. The teacher checks each table has the correct units.
Phase 2 Resources	- Pen drive - Hard Disk - a portable hard drive or picture of one - CD-ROM and DVD-ROM discs (or pictures) - chart of storage capacities and read/write speeds
Phase 3 Plenary	- Review the steps for Hard drives and optical storage media, discuss the common problem met and how it was solved, and link to everyday technology. Preview the next lesson ('Components of Computers and Computer Systems'). - Home Assignment: Find one storage device at home or school. Write the capacity printed on it, whether it is magnetic or optical, and one advantage of that type of storage.
Evaluation	Ask each group to name the storage item with the largest capacity on the table and give the evidence from the label. Check that learners can say why an optical disc is read-only while a hard drive can be written many times. Assessment mode: Oral questions and observation of participation.

Assignments

Class Assignment	Groups build a comparison table of the hard drive, portable hard drive, CD-ROM and DVD-ROM using the printed capacities and write speeds, then state one strength and one weakness of each. The teacher checks each table has the correct units.
Home Assignment	Find one storage device at home or school. Write the capacity printed on it, whether it is magnetic or optical, and one advantage of that type of storage.
Homework	Find one storage device at home or school. Write the capacity printed on it, whether it is magnetic or optical, and one advantage of that type of storage.

LESSON PLAN

Lesson Metadata

Week	4
Week Ending	02-10-2026
School	Achimota Basic School
Teacher	Persistence Teacher
Teaching Week	10
Lesson Date	02/10/2026
Duration (minutes)	60
Lesson Topic	Windows desktop: Start screen, tiles and taskbar

Curriculum Alignment

Strand	Introduction to Computing
Sub Strand	Components of Computers and Computer Systems
Content Standard	B7.1.1.2
Learning Indicator	B7.1.1.2.1
Performance Indicator	Learners can demonstrate Windows desktop: Start screen tiles and taskbar
Core Competencies	Digital Literacy, Communication and Collaboration
Key Words	Bootable Flash drive, Windows desktop, Start screen, tiles, taskbar, preview thumbnail, window, components, computers

WAEF Special Fields

Lesson Delivery

Phase 1 Starter	- This lesson builds on 'Components of Computers and Computer Systems' and moves the class on to Windows desktop: Start screen, tiles and taskbar. - Build on the previous lesson ('Components of Computers and Computer Systems'). Ask learners what they see when a Windows computer finishes starting. Record the parts they name on the board and add any missing ones.
Phase 2 Main	- Watch and listen carefully - then describe each step of Windows desktop: Start screen - tiles and taskbar in your own words - Work with your partner/group to carry out the task for Windows desktop: Start screen - tiles and taskbar and record what you find - At the computer (or using a screenshot on the board) point out the Start screen - the tiles - the taskbar buttons and a preview thumbnail. Learners name each part as the teacher points to it - In pairs at a computer or with the screenshot - learners open two windows and use the taskbar to preview one another - then write down the four features they used and what each one does - Demonstrate the concept or steps for Windows desktop: Start screen - tiles and taskbar. Use the board or an unplugged model if there are not enough devices for everyone - Class Assignment: Pairs use the computer (or the labelled screenshot) to open two windows, preview one from the taskbar, and record the four desktop features with what each one does. The teacher observes and asks each pair to demonstrate one feature.
Phase 2 Resources	- Bootable Flash drive - a computer with Windows - or a labelled screenshot of the desktop - chart naming the Start screen - tiles - taskbar and preview thumbnails
Phase 3 Plenary	- Review the steps for Windows desktop: Start screen, tiles and taskbar, discuss the common problem met and how it was solved, and link to everyday technology. Preview the next lesson ('Health and Safety in using ICT tools'). - Home Assignment: Draw the Windows desktop from memory and label the Start screen, tiles and the taskbar. Write one sentence on what a preview thumbnail shows.
Evaluation	Point to a screen part and ask a learner to name it and say what it does; require all four feature names across the class. Ask a pair to demonstrate previewing a window from the taskbar. Assessment mode: Oral questions and observation of participation.

Assignments

Class Assignment	Pairs use the computer (or the labelled screenshot) to open two windows, preview one from the taskbar, and record the four desktop features with what each one does. The teacher observes and asks each pair to demonstrate one feature.
Home Assignment	Draw the Windows desktop from memory and label the Start screen, tiles and the taskbar. Write one sentence on what a preview thumbnail shows.
Homework	Draw the Windows desktop from memory and label the Start screen, tiles and the taskbar. Write one sentence on what a preview thumbnail shows.

LESSON PLAN

Lesson Metadata

Week	6
Week Ending	16-10-2026
School	Achimota Basic School
Teacher	Persistence Teacher
Teaching Week	10
Lesson Date	16/10/2026
Duration (minutes)	60
Lesson Topic	Introduction to Computing - Health and Safety in using ICT tools

Curriculum Alignment

Strand	Introduction to Computing
Sub Strand	Health and Safety in using ICT tools
Content Standard	B7.1.2.1
Learning Indicator	B7.1.2.1.1
Performance Indicator	Learners can explore and talk about Health and Safety in using ICT tools
Core Competencies	Communication and Collaboration
Key Words	Smart Phone, Pictures, health, safety

WAEF Special Fields

Lesson Delivery

Phase 1 Starter	- This lesson builds on 'Components of Computers and Computer Systems' and moves the class on to Health and Safety in using ICT tools. - Build on the previous lesson ('Components of Computers and Computer Systems'). Recall the last ICT skill learned and connect it to Health and Safety in using ICT tools. If devices are limited, use an unplugged analogy from everyday life.
Phase 2 Main	- Watch and listen carefully - then describe each step of Health and Safety in using ICT tools in your own words - Complete the task for Health and Safety in using ICT tools on your own and check your work before you finish - Demonstrate the concept or steps for Health and Safety in using ICT tools. Use the board or an unplugged model if there are not enough devices for everyone - Learners practise Health and Safety in using ICT tools in small groups taking turns - or on paper/board using the same steps where devices are shared - Learners complete a short task applying Health and Safety in using ICT tools on their own and troubleshoot one common problem - Class Assignment: In groups, learners demonstrate the task for Health and Safety in using ICT tools while the teacher observes and records which learners have met the indicator.
Phase 2 Resources	- Smart Phone - Pictures
Phase 3 Plenary	- Review the steps for Health and Safety in using ICT tools, discuss the common problem met and how it was solved, and link to everyday technology. - Home Assignment: Practise the steps for Health and Safety in using ICT tools at home and be ready to show one step in the next lesson.
Evaluation	Observe learners performing Health and Safety in using ICT tools against a steps checklist, or mark a paper-based version where devices are not available. Assessment mode: Oral questions and observation of participation.

Assignments

Class Assignment	In groups, learners demonstrate the task for Health and Safety in using ICT tools while the teacher observes and records which learners have met the indicator.
Home Assignment	Practise the steps for Health and Safety in using ICT tools at home and be ready to show one step in the next lesson.
Homework	Practise the steps for Health and Safety in using ICT tools at home and be ready to show one step in the next lesson.